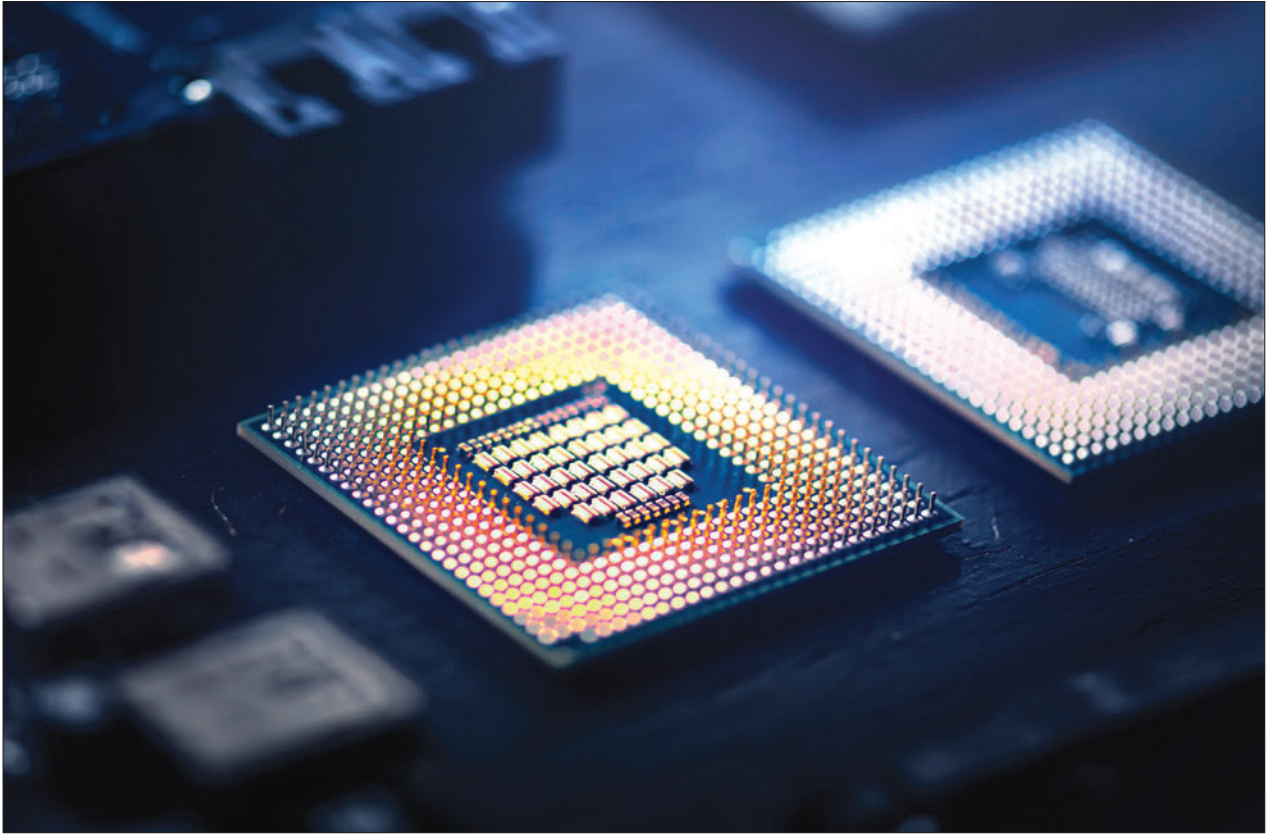




The Beating Heart – CPUs

All about choosing the right CPUs for your 2025 PC build

A personal computer's central processing unit (CPU) is often called its "beating heart." It orchestrates every calculation, controls how fast your programs run and determines what kinds of tasks your system can handle.

When building your own PC, understanding CPUs is crucial because the processor governs everything from gaming frame rates to how quickly you can compile code or edit videos. In 2025, the choice has never been more nuanced: both AMD and Intel have released new chip families that blend traditional compute cores with AI acceleration, new memory architectures and highly efficient manufacturing processes. This chapter

demystifies how modern CPUs work, compares AMD and Intel's latest offerings and guides you step-by-step through choosing and installing the right chip for your needs.

Anatomy of a Processor

A CPU is a tiny piece of silicon that contains billions of transistors. Those transistors form cores, the basic units of computation. A core executes instructions, decodes them, performs arithmetic and interacts with memory. Modern CPUs typically have multiple cores so they can process several tasks simultaneously. Each core may support threads, which allow the operating system to schedule two instruction streams on the same core using